Eurographics Workshop on Visual Computing for Biology and Medicine (VCBM) (2026)
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Interactive Visualization of Dynamic Functional Connectivity in the ABIDE Autism Dataset

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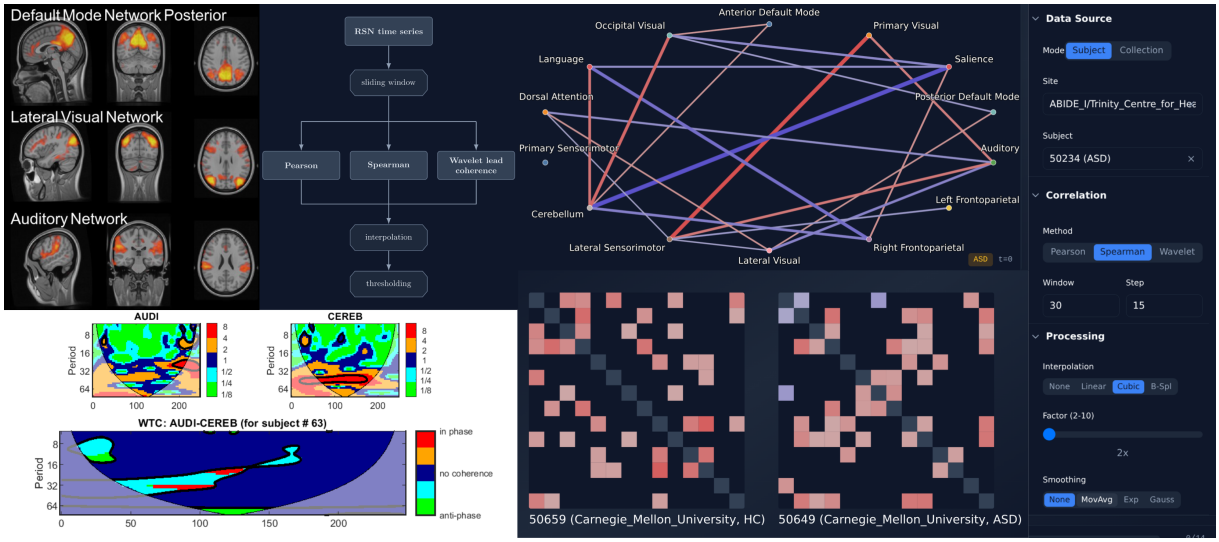


Figure 1: ***Networks (left):*** *example resting-state networks (RSNs) from independent component analysis.* ***Processing (center):*** *each RSN time series is correlated over a sliding window using Pearson, Spearman, or wavelet lead coherence, then interpolated and thresholded.* ***Wavelet coherence (lower left):*** *wavelet lead coherence is derived from wavelet transform coherence (WTC) maps of network pairs, classified as lead, lag, in-phase, or anti-phase coherence.* ***Tool (right):*** *graph and overview panes for comparing individual subjects as well as the healthy control (HC) and autism (ASD) diagnostic classes.*

Abstract

*Autism spectrum disorder (ASD) is currently diagnosed using observation-based criteria, while an objective test is desired [SPAZ24]. Resting state functional connectivity between brain regions is increasingly understood to be a dynamic process, varying in time [HWA*13, PBVDV16]. However, the tools currently available collapse a recording into a single correlation matrix and discard the time variations. We present an interactive web application for exploring the time-varying functional connectivity between brain regions in the large ABIDE I+II autism fMRI dataset. Each recording is decomposed into resting state networks whose dynamic connectivity is measured over a sliding window using several methods, including wavelet lead coherence, a directional measure of which network leads which. These connections are visualized as a video of a graph with weighted edges, alongside an overview pane to compare individual scans or entire diagnostic classes. Our system turns a large, otherwise opaque dataset into a visual format for discovering time-varying functional connectivity patterns. It is a step toward an interpretable neuroimaging tool that may aid diagnosis.*

CCS Concepts

• *Information systems* → Information systems applications → Decision support systems → Data analytics

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1. Introduction

Resting state fMRI (rs-fMRI) is the most studied imaging method in autism spectrum disorder (ASD) research [SPAZ24], and the functional connectivity it reveals is central to studying the condition. This connectivity is conventionally computed over an entire recording, resulting in a single static connectivity matrix. However, over the course of a recording, connectivity patterns can vary as the mind wanders. Dynamic functional connectivity captures the information that static correlations discard [CG09], and has shown promise as a biomarker for the diagnosis of autism [BAZ17], depression [CPB*23, PLS*24], and cognitive aging [BBAZ21]. As objective tests are meant to support clinical decisions, their value is subject to the trust of clinicians. Therefore, a visualization of dynamic connectivity patterns can help clinicians understand a diagnostic feature, rather than trust an opaque prediction.

The Autism Brain Imaging Data Exchange (ABIDE) is a large, comprehensive, multisite rs-fMRI ASD dataset [SPAZ24]. Recent work preprocessed ABIDE I+II using independent component analysis (ICA) into resting state networks (RSNs) for 900 subjects (417 ASD, 483 controls) [SPA*25]. This preprocessed data is well suited to studying dynamic functional connectivity patterns, yet common visualizers are static and lack the interactive, time-varying exploration needed to observe them. We present a novel web application that fills this gap with an animated graph view of windowed connectivity between 14 RSNs, computed per subject from configurable parameters. It computes this connectivity using symmetric correlations and a directional wavelet lead coherence measure, and provides a comparative overview across diagnostic classes and specific sites. Figure 1 shows an overview of the system, from the RSNs to the interactive graph and overview views.

2. Neuroimaging and Brain Connectivity

From group ICA, 14 resting state networks were identified [SPA*25]. Functional connectivity between two networks is quantified by correlating their time series. Applied over a fixed-length sliding window, this produces a series of matrices that describe the dynamic functional connectivity [HWA*13, PBVDV16].

The system offers three correlation measures. Pearson and Spearman correlation produce symmetric matrices in the range $[-1, 1]$. The third is a directional (asymmetric) measure, called wavelet lead coherence. The wavelet transform decomposes each signal into time and frequency, capturing how its frequency content varies over time rather than averaging over the whole scan. Wavelet coherence then acts as a localized correlation coefficient between two signals, where the cross-wavelet's phase angle contains the relative phase difference between the signals [GMJ04, TC98, CG09]. This information is used to segment the coherence map into lead, lag, in-phase, or anti-phase coherence of one signal with respect to the other [BAZ17, CPB*23]. The wavelet lead coherence measure reports values in the range $[0, 1]$, the fraction of time and frequency points at which one network leads the other, computed in both directions, so the resulting matrix is asymmetric. This directional measure can also reflect causal influences, which may motivate further research into causal relationships between networks [CPB*23].

3. Visualizations of Dynamic Brain Activity

The graph view places the resting state networks on a ring and draws the current frame's connectivity matrix as edges (Figure 1). The color of an edge encodes the correlation value on a blue to red scale and thickness encodes its absolute magnitude. A configurable threshold is used to hide weak edges and denoise the visualization. For wavelet lead coherence, the edges carry arrows showing the directionality of the correlation. The frames are animated over time, showing how the graph changes. Hovering over a network node isolates its edges, and hovering over an edge shows the two networks it connects and their correlation value. This explorative view can also play aggregate sequences, the mean or median over all ASD or all control subjects, so an individual's dynamics can be compared against either class. Any change in the parameters (window size, step length, correlation measure, smoothing, interpolation, and thresholding) recomputes the graph in real time, enabling users to explore dynamic connectivity patterns quickly.

A complementary overview pane (Figure 1) reduces each subject to a single correlation matrix over the entire recording, shown as a heatmap or a miniature graph, separated into neurotypical control and ASD columns and grouped by acquisition site. Read against the class mean and median, the overview enables fast scanning for individual outliers, group contrasts, and site variation. Identifying site variation and measurement error induced outliers is especially important, as it is a known confound in multisite rs-fMRI [SPAZ24].

4. Conclusions

Dynamic functional connectivity is often studied as a diagnostic marker, yet its high-dimensional nature limits interpretability, and therefore its utility for the clinicians these tools are meant to support. Existing tools visualize dynamic connectivity statically, so the dynamic patterns are difficult to observe. Our contribution is a novel visualization tool for observing dynamic functional connectivity patterns between RSNs, applied to the ABIDE I+II dataset and publicly available at `https://brainviz.ideabattery.org/`. This is a step toward translating high-dimensional fMRI features into interpretable information that can aid clinicians, which can help explain the features important to diagnostic tools. Planned work adds a quantitative comparison of ASD against control subjects, turning exploratory observations into testable group differences and diagnostic features.

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